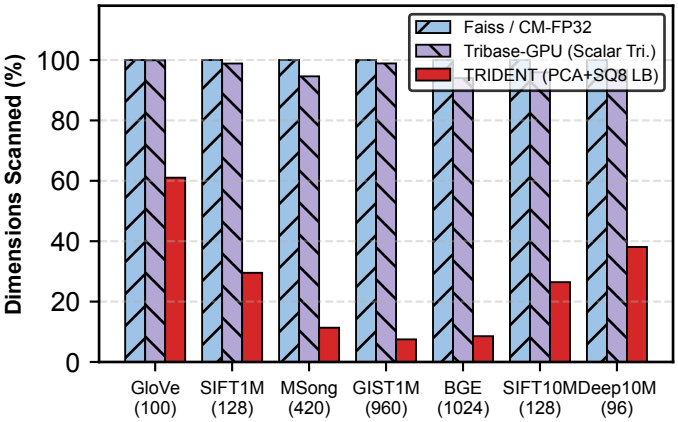
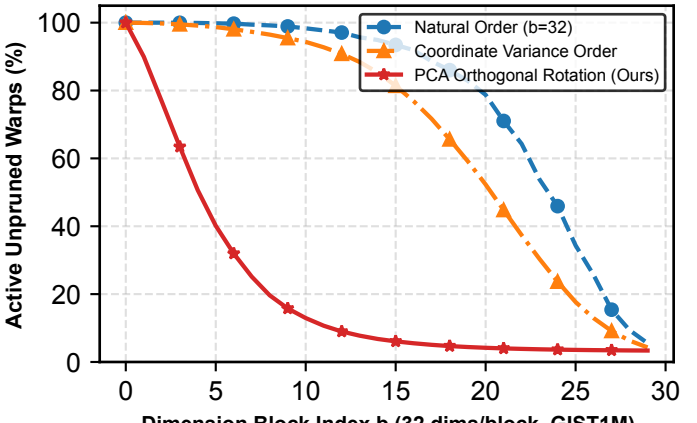

(a) Obs. 1: Scalar Bounds Collapse in High-D
Dimensions Scanned (%)
0
20
40
60
80
100
Faiss / CM-FP32
Tribase-GPU (Scalar Tri.)
TRIDENT (PCA+SQ8 LB)
GloVe (100)
SIFT1M (128)
MSong (420)
GIST1M (960)
BGE (1024)
SIFT10M (128)
Deep10M (96)


(b) Obs. 2: PCA Energy Compaction Accelerates Pruning
Active Unpruned Warps (%)
0
20
40
60
80
100
Natural Order (b=32)
Coordinate Variance Order
PCA Orthogonal Rotation (Ours)
0
5
10
15
20
25
30
Dimension Block Index b (32 dims/block, GIST1M)